

1. Administrative & Study Identification

Full Study Title: An Interventional Phase 2/3, Adaptive, Multi-center, Randomized, Double-blind Study to Investigate Efficacy and Safety of Oral Sisunatovir Compared with Placebo in Non-Hospitalized Symptomatic Adults with Respiratory Syncytial Virus Infection Who are at Risk of Progression to Severe Illness **Short Title/Acronym:** A Study to Learn About the Study Medicine Sisunatovir in Adults With Respiratory Syncytial Virus (RSV) Infection **Protocol Number:** C5241007 **NCT Number:** NCT06079320 **Sponsor Name:** Pfizer **Investigational Product:** Sisunatovir (RSV F-protein inhibitor) **Phase of Development:** Phase 2/Phase 3 **Medical Monitor:** Pfizer Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the efficacy and safety of oral sisunatovir compared with placebo in preventing severe clinical outcomes (specifically, RSV-related hospitalization or death from any cause through Day 28) in non-hospitalized symptomatic adults with RSV infection who are at risk of progression to severe illness. **Secondary Objectives:** To evaluate the proportion of participants with progression, development, or resolution of Lower Respiratory Tract Infection (LRTI) through Day 10 and Day 15, and the mean number of days alive and free from hospital stays.

2.2 Background & Rationale Respiratory Syncytial Virus (RSV) is a viral pathogen that causes acute respiratory infections and can lead to severe illness, such as bronchiolitis or pneumonia, particularly in high-risk individuals. Sisunatovir is an oral RSV F-protein inhibitor designed to prevent the virus from fusing with host cells. This study was initiated to determine if an early 5-day course of sisunatovir could prevent disease progression and hospitalization in high-risk, non-hospitalized adults. *(Note: The study was terminated early due to business reasons and development challenges, including a drug-drug interaction with antacids, rather than safety concerns with the drug itself).*

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Placebo-controlled **Blinding:** Double-blind (Quadruple: Participant, Care Provider, Investigator, Outcomes Assessor) **Randomization:** Randomized (Two parallel-arm, adaptive)

3.2 Planned Enrollment & Duration Target **\$N\$**: Originally planned for ~2,715 participants; Actual Enrollment was 16 due to early termination. **Number of Sites**: Multi-center (Global trial including sites in the US, Japan, Canada, Australia, etc.) **Estimated Study Start**: October 2023 **Estimated Primary Completion**: Terminated (Originally projected for late 2025)

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Participants aged 18 years or older at screening. 2. Diagnosis of RSV infection collected within 5 days prior to randomization. 3. New onset or worsening of at least one of the following signs and/or symptoms consistent with a viral acute respiratory infection (ARI) within 5 days prior to randomization: nasal congestion, nasal discharge, sore throat, cough, sputum production, shortness of breath, or wheezing. 4. Has at least 1 risk factor for severe illness: ≥ 65 years of age, chronic lung disease, heart failure, or an immunosuppressive condition/medication.

4.2 Exclusion Criteria 1. Any medical or psychiatric condition that may increase the risk of study participation or interfere with study evaluation. 2. Diagnosis of viral respiratory infections other than RSV, including influenza and SARS-CoV-2. 3. Current need for hospitalization or anticipated need for hospitalization for any reason within 24 hours after randomization. 4. Any clinically significant ECG abnormality in the pre-dose ECG. 5. Hypersensitivity or other contraindication to any components of the study interventions.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Sisunatovir oral tablets or matching placebo. **Route of Administration**: Oral **Duration of Treatment**: 5 days

5.2 Concomitant & Prohibited Therapy Permitted: Standard maintenance medications for underlying co-morbidities (e.g., heart failure, chronic lung disease). **Prohibited**: Concomitant use of specific antacids (due to identified drug-drug interactions affecting absorption) and other investigational antiviral agents.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
1	Screening/Baseline	Day 1 Informed Consent, RSV testing, ECG, Randomization, First Dose
2	Treatment Phase	Days 1 - 5 Daily dosing of oral Sisunatovir or Placebo, Symptom Tracking
3	Follow-up / Interim	Days 10, 15 Clinic visits/ nose swabs, Blood work, LRTI progression/resolution assessment

| **End of Study** | Day 28 (approx. Week 5) | Final evaluation of hospitalization status, mortality, and safety follow-up |

(Note: Participants were expected to attend 8-10 study clinic visits/contacts over the 5-week study period).

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Proportion of participants with RSV-related hospitalization or death from any cause through Day 28 (specifically evaluated among patients treated ≤ 3 days after RSV symptom onset). **Measurement Method:** Hospital admission records and mortality tracking evaluated by the investigator.

7.2 Secondary & Exploratory Endpoints * Proportion of participants with progression or development of Lower Respiratory Tract Infection (LRTI) through Day 10. * Proportion of participants with resolution of LRTI at Day 15. * Mean number of days alive and free from hospital stay (hospital-free days) through Day 28.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Conducted via clinic visits, questionnaires, follow-up phone calls, and routine blood work throughout the 5 weeks. **Serious Adverse Events (SAEs):** Continuously monitored up to Day 28. (Note: Upon trial termination, the sponsor noted there were no safety concerns in the decision to stop the study). **Data Monitoring Committee (DMC):** Employed to oversee the adaptive design aspects of the Phase 2/3 trial.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) for primary efficacy evaluation; Safety Set for AEs. **Sample Size Justification:** Originally targeting $\geq 2,700$ subjects to adequately power a superiority assessment comparing sisunatovir versus placebo in reducing the rare events of hospitalization and death. **Handling of Missing Data:** Standard multiple imputation for missing symptom diaries and central PCR viral load data (undetectable viral load analyzed as 0 copies/mL).

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study was conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki.

Monitoring Plan: Centralized monitoring of PCR laboratory results

(target not detected/TND definitions) and site-level review of informed consent and ARI symptom capture. **Audit Trail:** All patient-reported outcomes (questionnaires) and ECG data were captured via validated electronic systems maintaining full audit trails.

11. Study Identifiers & Governance

Primary ID: C5241007 **Registry Number:** NCT06079320 **Sponsor/Lead Organization:** Pfizer **Study Title:** A Study to Learn About the Study Medicine Sisunatovir in Adults With Respiratory Syncytial Virus (RSV) Infection **Official Regulatory Title:** An Interventional Phase 2/3, Adaptive, Multi-center, Randomized, Double-blind Study to Investigate Efficacy and Safety of Oral Sisunatovir Compared with Placebo in Non-Hospitalized Symptomatic Adults with Respiratory Syncytial Virus Infection Who are at Risk of Progression to Severe Illness

12. Clinical Framework (RSV Specific)

Primary Condition: Respiratory Syncytial Virus (RSV) Infection **Diagnosis Threshold:** PCR or rapid test confirmed RSV ≤ 5 days prior to randomization with ≥ 1 ARI symptom. **Medical Methodology:** Direct-acting antiviral (F-protein inhibitor) therapy. **Optimization Target:** Prevention of progression to severe Lower Respiratory Tract Infection (LRTI) and hospitalization.

13. Study Procedures & Methodology

Management Pathway: Early outpatient intervention (within 5 days of symptom onset) targeting viral suppression. **Education Protocol (HCP):** Investigators instructed on precise definitions of LRTI transition (e.g., ≥ 2 lower respiratory signs/symptoms for at least 24 hours). **Education Protocol (Patient):** Counseling on exact 5-day dosing adherence, symptom diary questionnaires, and avoiding specific contraindicated antacids. **Quality Audit Mechanism:** Central laboratory confirmation of RSV viral load and standard clinical response evaluations.

14. Observation & Follow-Up Schedule

Total Duration: 5 Weeks (28 Days) **Onsite Visit Cadence:** Frequent visits centered around Days 1, 5, 10, 15, and 28 (approx. 8-10 interactions total). **Remote Monitoring Frequency:** Supplemental follow-up phone calls. **Checklist Review Interval:** Daily questionnaires capturing nasal congestion, cough, and dyspnea progression/resolution.

15. Sign-off & Approval

Role	Name	Signature	Date		:---	:---	:---	:---
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]					
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]					Lead
Statistician	[Name]	_____	[DD-MMM-YYYY]					